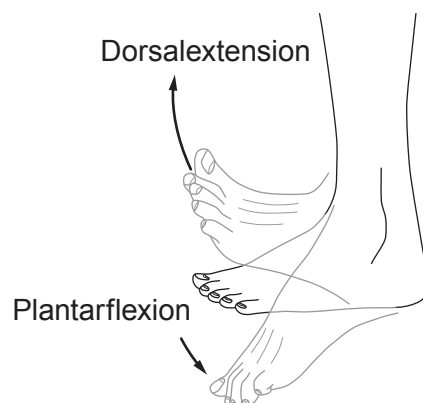


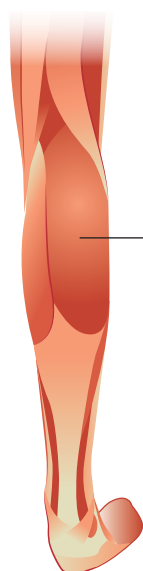
3. Healthy muscles and joints are essential for pain-free movement. These are very important for successful athletes.

- (a) The ankle is a type of hinge joint that is used when we walk and run. Its action is shown in the diagram below.



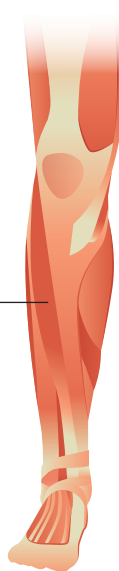
This movement is controlled by an antagonistic pair of muscles called the calf and tibialis anterior. The calf is at the back of the leg and the tibialis anterior is at the front of the leg.

Back of leg



Calf
muscle

Front of leg



Tibialis
anterior
muscle

Explain how this pair of muscles produce dorsalextension and plantarflexion.

[3]

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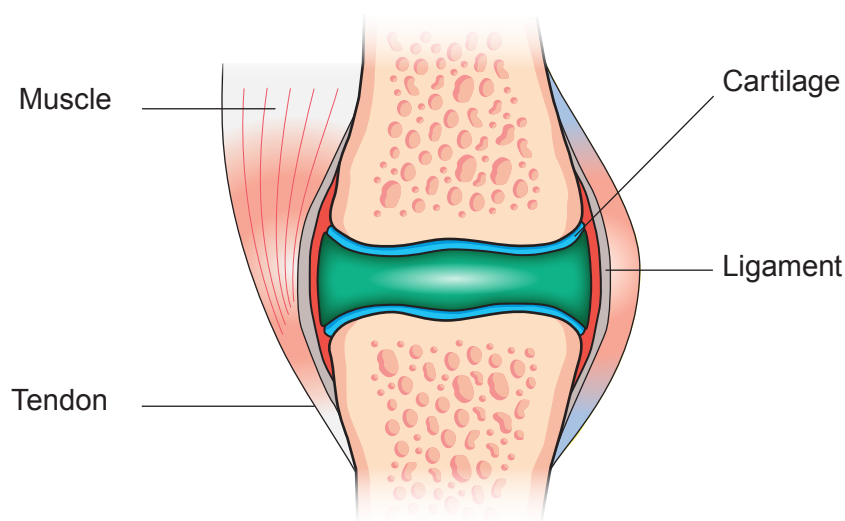
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- (b) The knee is a synovial hinge joint.



Explain how damage to cartilage and ligaments may affect an athlete.

[3]

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- (c) Two athletes, Dafydd and Tom, compete in a 400 m training race and their finish times are identical. Their times to travel certain distances within the race are measured and are shown in the table below.

Distance (m)	0	50	100	150	200	300	400
Tom's time (s)	0	7.65	13.65	20.48	27.30	41.93	56.55
Dafydd's time (s)	0	6.10	11.40	18.40	25.40	40.95	56.55

- (i) Dafydd thought his fastest speed was for the last 100 m of the race. Explain whether you agree.

[3]

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- (ii) Since they finished the race in the same time, Tom thought their maximum speed must have been the same. Plot graphs of distance against time for Tom and Dafydd and explain whether you agree with Tom. [5]



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